Reviewer Pack — Minimal Hyperbolic Rank and Resolution Proof Width Bound

Entry 8387fdbf6f65 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Hyperbolic Rank and Resolution Proof Width Bound

Entry ID: `8387fdbf6f65` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-06-08 03:23:44 UTC

1. Conjecture

Field A (mathematical branch): Hyperbolic Geometry (Gromov-Hausdorff Distance) **Field B** (complexity object): Boolean Satisfiability (Resolution Proof Complexity)

Statement:

> For any given Boolean satisfiability instance φ, the resolution proof width of φ is bounded by a function γ(n) that depends only on the number of variables n and not on the structure of φ, such that w(φ) ≤ γ(n). The function γ(n) can be computed in polynomial time using the Gromov-Hausdorff distance between φ's clause indicator graph and its hyperbolic realization.

Rationale (proposer's reasoning):

> Hyperbolic geometry has been successfully applied to model complex systems with intricate structures, such as networks. By leveraging the Gromov-Hausdorff distance, we can compare the structure of the clause indicator graph of a satisfiability instance to a hyperbolic realization, which may expose underlying geometric properties related to resolution proof width.

Taxonomy category: `Hyperbolic_Geometry_×_Boolean_Satisfiability` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `dc745bb5349a2192`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if the Gromov-Hausdorff distance between the clause indicator graph and its hyperbolic realization is computable in polynomial time and all instances have a resolution proof width ≤ γ(n) for computed γ(n).

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Gromov-Hausdorff distance" AND "resolution proof complexity" - "Hyperbolic geometry" AND "Boolean satisfiability resolution" - "clause indicator graph" AND hyperbolic realizations

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=124, elapsed=240.6s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_random_boolean_instance(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def gromov_hausdorff_distance(graph1, graph2):
        # Placeholder implementation
        return 1.0

    def resolution_proof_width(instance):
        # Placeholder implementation
        return len(instance)

    n = random.choice([5, 10, 15, 20, 30, 40])
    instance = generate_random_boolean_instance(n)
    distance = gromov_hausdorff_distance(instance, instance)  # Placeholder
    width = resolution_proof_width(instance)

    return {
        "metric_name": "resolution_proof_width",
        "metric_value": width,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19,
```

```
results = []
for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_value = sum(r["metric_value"] for r in results) / len(results)
std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results)
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
else:
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_fai
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
TIMEOUT after 240s
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> The test timed out before producing data, which means we cannot confirm whether the Gromov-Hausdorff distance computation is in P-time or if all instances have a resolution proof width ≤ γ(n). | next: Re-run the test with increased time limits to ensure it completes and verify the computability of the Gromov-Hausdorff distance in polynomial time.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14658
2	propose	ollama_remote	glm4:latest	0	0	18066
3	preregistration	ollama_remote	glm4:latest	0	0	9474
4	novelty	ollama_remote	glm4:latest	0	0	11393
5	novelty	ollama_remote	glm4:latest	0	0	8693
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14144

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11046
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8494
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6280
10	judge	ollama_remote	glm4:latest	0	0	16732

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 118980 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/8387fdbf6f65.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/8387fdbf6f65.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/8387fdbf6f65.tar.gz` (if generated)